CHAPTER 1

Sampling Data

1.1 Planning a Study

Transit routes, school schedules, and drug approvals all rest on data, and each decision is only as good as the data behind it. A study plan states what question you are answering, how you will gather information, and whom you will gather it from.

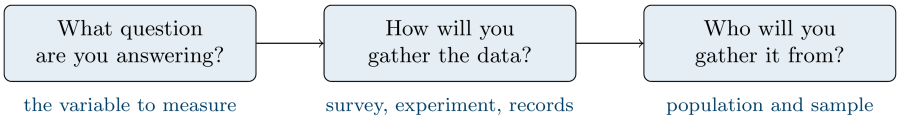


Figure 1.1: A study plan answers three questions: what you want to know, how you will collect data, and whom you will collect it from.

1.1.1 Vocabulary

population The entire group of individuals or items that you want to learn about. The population is defined by the question you are asking.

frame Also called a sampling frame. A list of all the members of the population from which a sample will be drawn. The frame and the population often differ slightly. A school's enrollment database, a city's voter registration list, and a company's customer records are all examples of frames.

sample The part of the population that is actually studied. You collect data from the sample and use it to draw conclusions about the population.

sample survey The process of collecting information from a sample. The information gathered is then used to make inferences about population parameters.

census The process of collecting information from every member of the population rather than just a sample. A census gives complete information, but it comes at a cost.

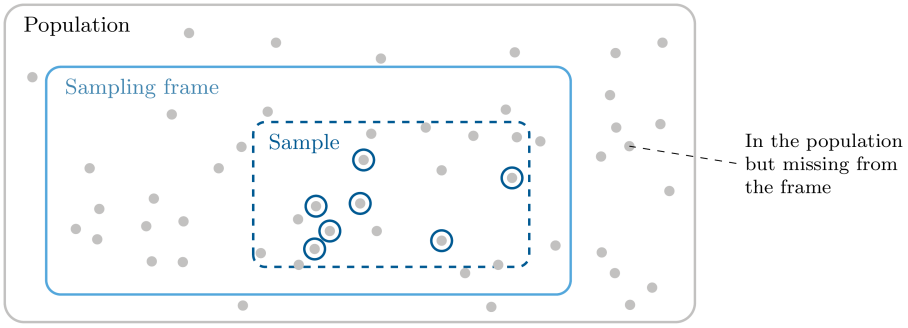


Figure 1.2: The sample is drawn from the sampling frame, which is a working list of the population. Members missing from the frame can never be sampled.

Exercise 1 Identifying the Population and Sample

A city's parks department wants to understand how often residents use public parks. Staff randomly select 200 households from the city's water billing records and conduct phone interviews.

Working Space

1. What is the population?
2. What is the frame?
3. What is the sample?
4. Is this a census? Explain.

Answer on Page 17

1.1.2 Why Not Always Use a Census?

A census measures everyone in the population. For a small group, such as a football team, that is easy.

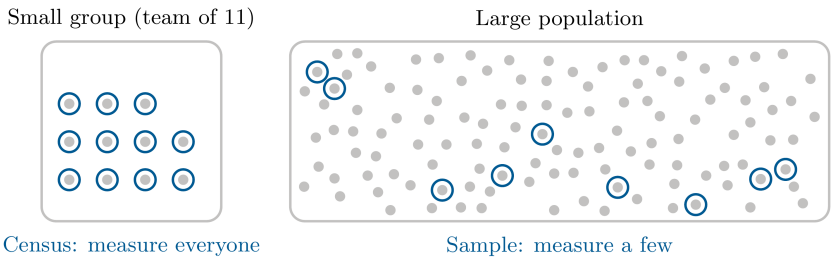


Figure 1.3: A census measures every member of a small group. For a large population, a sample of members is measured instead.

For most populations, a census is impractical:

- The population is too large. A company that wants to know what all households watch cannot contact every one.
- The measurement destroys the item. If testing a battery drains it, checking every battery would leave none to sell.
- The information goes out of date. If a survey of everyone takes months, opinions may have changed by the time it ends.

Exercise 2 Census or Sample?

Working Space

For each situation, decide whether a census is feasible or whether a sample should be used. Briefly explain your reasoning.

Situation	Census or Sample? Explain
A school librarian wants to know which of her 8 student volunteers prefers morning or afternoon shifts.	
A battery manufacturer wants to test the lifespan of batteries by running them until they die.	
A government agency wants to estimate the average commute time of workers across a country of 40 million employed adults.	
A restaurant owner wants to know the dietary restrictions of the 22 guests attending a private event.	

Answer on Page 17

1.2 Planning and Conducting Surveys

The way you choose people determines whether the sample represents the population or leaves parts of it out.

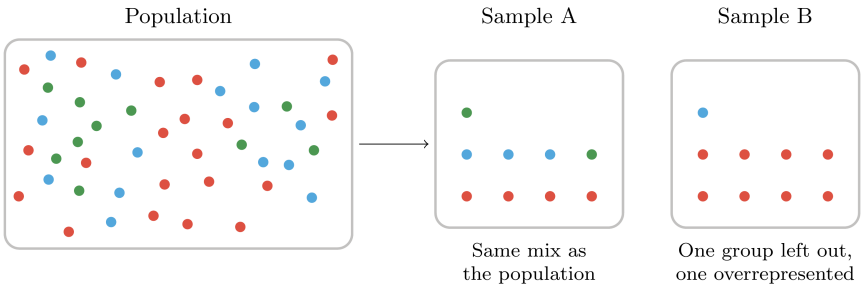


Figure 1.4: Sample A keeps the same mix of groups as the population. Sample B leaves one group out and overrepresents another.

1.2.1 Biased Sampling

A sample is *biased* if it favors some people or answers over others. Bias does not go away just because the sample is bigger.

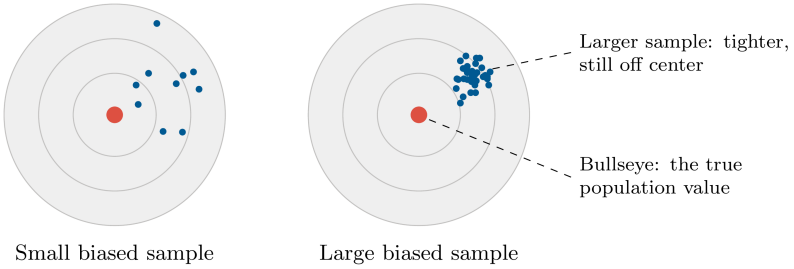


Figure 1.5: A larger biased sample clusters more tightly, but its center stays away from the true population value.

Each of the following methods selects people without chance, so each tends to produce a biased sample.

With *judgmental sampling*, one person picks whoever they believe looks representative. A school inspector who visits only the classrooms she already rates highly is using judgmental sampling.

A *convenience sample* uses whoever is easiest to reach. A journalist who interviews only people at the bus stop outside his office will miss commuters who drive, cycle, or travel at other hours.

When people decide for themselves whether to answer, the result is a *voluntary response sample*. An online poll on a news site attracts mostly visitors with strong opinions, while people who care less rarely respond.

Researchers using *quota sampling* recruit people until a preset count for each group, such as an age band or gender, is filled. If a population is 60% female and 40% male, a quota sample of 100 stops recruiting women at 60 and men at 40. The group mix matches the population, but whoever is approached first fills each quota.

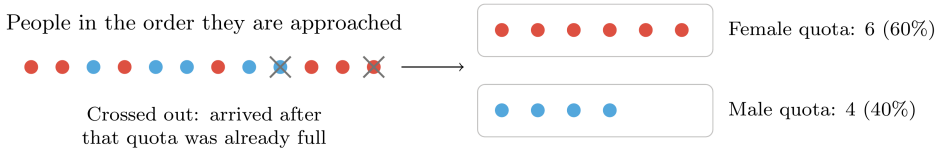


Figure 1.6: Quota sampling fills each group to a fixed number in the order people are approached. Selection within each group is not random.

For populations that are hard to reach, *snowball sampling* starts with a few qualifying participants and asks each of them to refer others who also qualify. Researchers use it to study hidden groups, such as people experiencing homelessness or people with a rare disease, but the sample follows social networks rather than chance. *Snowball sampling is beyond the scope of the AP Statistics exam.*

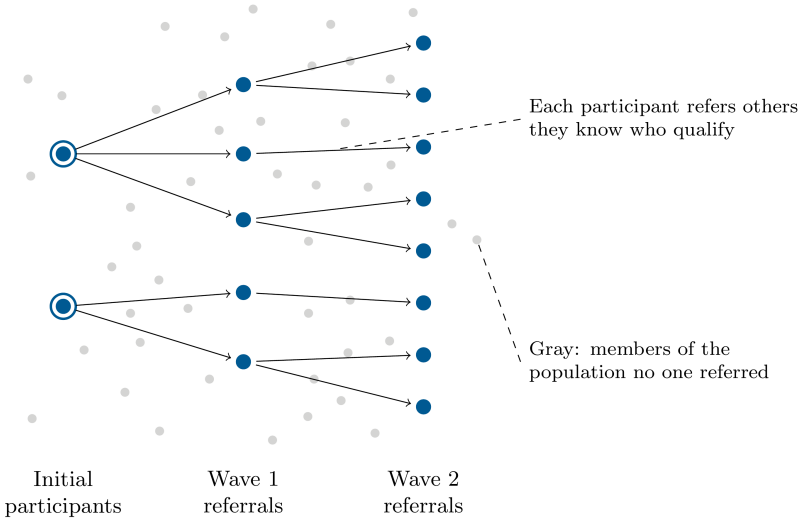


Figure 1.7: Snowball sampling starts with a few initial participants. Each wave of referrals recruits the next, so the sample grows through personal connections.

All five methods leave selection to judgment or convenience. Random selection gives every member a fair chance of being chosen and usually produces more trustworthy results.

1.3 Simple Random Sampling

A *simple random sample* (SRS) is a sample chosen so every group of the same size has the same chance of being picked. An SRS can be drawn in two ways:

- In *sampling with replacement*, each chosen member returns to the pool and could be picked again.
- Each member can be chosen only once in *sampling without replacement*.

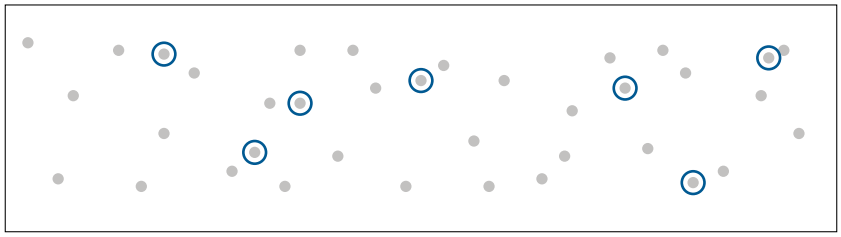


Figure 1.8: In simple random sampling, every individual in the population has an equal chance of being selected, and there is no grouping or pattern to who is chosen.

Chance, rather than convenience or judgment, must decide who is selected. Three ways to do this:

- Assign every member of the population a number. Use a random number table, entering at any random point and reading numbers sequentially. Select members whose numbers appear, skipping values outside your range and any repeats, until you have the sample size you need.
- Use a calculator. On a TI-84, `MATH` → `PRB` → `5:randInt(` generates random integers. The command `randInt(1,500,30)` produces 30 random integers between 1 and 500. Ignore repeats.
- Use a physical randomization device. Write each member's identifier on an identical slip of paper, mix thoroughly in a container, and draw without looking until the sample is complete.

Exercise 3 Selecting an SRS

A regional transit authority wants to survey 25 of its 300 bus drivers about workplace safety conditions.

1. Describe a complete procedure for selecting an SRS of 25 drivers using a random number table.
2. A supervisor suggests selecting the 25 drivers who are easiest to reach during the next shift change. What type of sample is this, and what bias might it introduce?
3. A colleague suggests selecting 5 drivers at random from each of the authority's 5 depots. Is this an SRS? Explain.

Working Space

Answer on Page 17

Computing in Python

In this exercise you will load real world population data published by the World Bank, treat all 265 countries and territories as your population, draw a simple random sample, and compare your sample mean to the true population mean.

1. Import the necessary libraries.

```
import pandas as pd
```

2. Load the dataset directly from the web and filter to the year 2020.

```
url =
 ↪  "https://raw.githubusercontent.com/datasets/population/main/data/population.csv"
df = pd.read_csv(url)

pop2020 = df[df["Year"] == 2020][["Country Name",
 ↪  "Value"]].reset_index(drop=True)
print(pop2020.head())
```

```
print(f"Total entries: {len(pop2020)}")
```

3. Calculate the true population mean. This is the parameter you are trying to estimate.

```
true_mean = pop2020["Value"].mean()
print(f"True population mean: {true_mean:,.0f}")
```

4. Draw an SRS of 20 countries and compute the sample mean.

```
sample = pop2020.sample(n=20)
sample_mean = sample["Value"].mean()
print(f"Sample mean (n=20): {sample_mean:,.0f}")
```

Exercise 4 Sampling Real Data from the Web

Working Space

1. Run step 4 five times and record the sample mean each time. How close is each to the true mean? Are they all the same?
2. Repeat question 1 with a sample size of 80. How do the results compare? What does this tell you about the relationship between sample size and sampling error?

Answer on Page 18

1.4 Other Random Sampling Methods

An SRS can be expensive or imprecise for some populations. Three other random methods appear regularly on the AP exam.

In *systematic sampling*, you list the population, choose a random start, and then pick every kth member. To sample 40 students from 800, start at a random number between 1 and 20, then take every 20th student. The method needs a complete list of the population.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60

Random start: 7

Then every 20th

7, 27, 47, ...

Figure 1.9: Systematic sampling with a random start of 7 and $k = 20$ selects members 7, 27, 47, and so on.

For *stratified random sampling*, you split the population into groups called strata whose members share a characteristic, such as students in the same grade or patients in the same ward. You then take a random sample from each stratum. Every group is guaranteed a place in the sample, which matters whenever groups need to be compared.

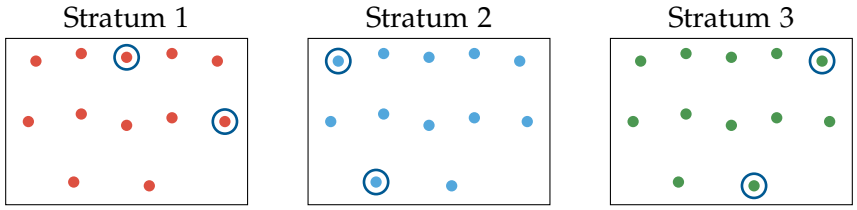


Figure 1.10: Each stratum is a single color, since members within a stratum share a characteristic. A small random sample (circled) is drawn from each stratum.

With *cluster sampling*, you split the population into groups called clusters, each with roughly the same mix of members as the whole population. You then randomly select some clusters and study everyone inside them. This suits populations spread over a wide area, where reaching everyone is expensive. A company surveying customers in 12 cities might randomly select 8 of them and survey every customer in each.

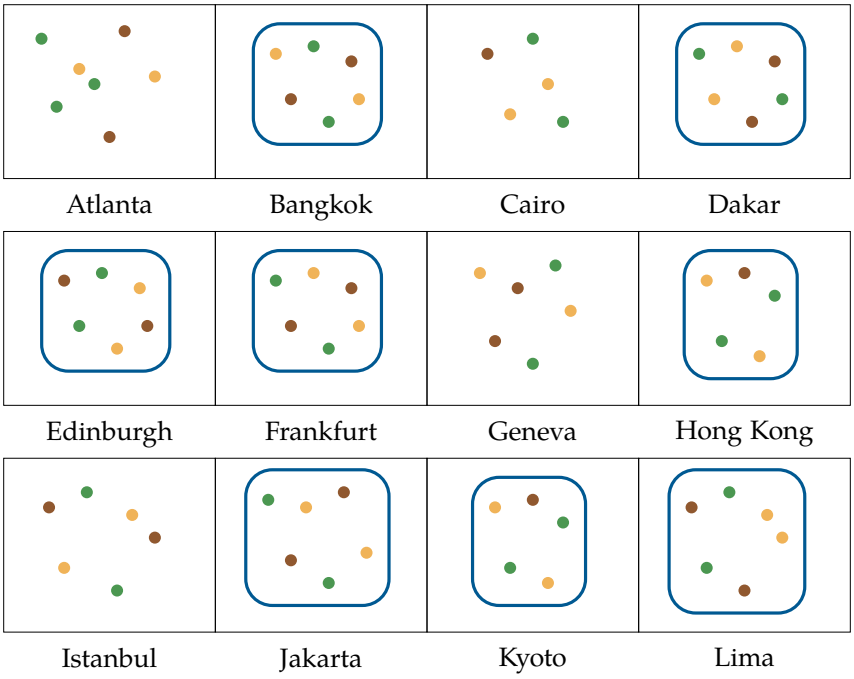


Figure 1.11: Each dot represents a customer and each color represents a different customer segment. Atlanta, Cairo, Geneva, and Istanbul were not selected. The other 8 cities were randomly selected as clusters, and every customer within each selected city becomes part of the sample.

Strata and clusters are both groups, but they play opposite roles:

strata Groups that differ from each other and are similar within. Take a random sample from every stratum.

clusters Groups that each resemble the whole population. Randomly select some clusters and include every member of each.

Exercise 5 Identifying Sampling Methods

Working Space

Identify the sampling method used in each scenario: SRS, systematic, stratified, or cluster.

Scenario	Method
A supermarket chain selects 10 of its 80 stores at random and surveys every employee in those stores.	
A university randomly selects 50 students from each of its four faculties to participate in a wellbeing survey.	
A water utility selects a random starting meter on its numbered list and then tests every 25th meter for contamination.	
A film festival puts all 900 registered attendee names in a database and uses software to randomly select 90.	

Answer on Page 18

Computing in Python

Apply three sampling methods to the same dataset and compare what each one produces. The dataset contains 30 students across three grade levels with their test scores.

Create the dataset.

```
import pandas as pd
import numpy as np

np.random.seed(42)

data = {
    "student_id": range(1, 31),
    "grade": ["9th"]*10 + ["10th"]*10 + ["11th"]*10,
    "score": (
        list(np.random.randint(60, 80, 10)) +
        list(np.random.randint(70, 90, 10)) +
        list(np.random.randint(80, 100, 10))
    )
}

```

```
df = pd.DataFrame(data)
print(df)
```

Compute the true population mean.

```
true_mean = df["score"].mean()
print(f"True mean score: {true_mean:.1f}")
```

1. Simple random sample. Draw an SRS of 9 students.

```
srs = df.sample(n=9)
print(f"SRS mean: {srs['score'].mean():.1f}")
print(srs[["student_id", "grade", "score"]])
```

2. Stratified random sample. Draw 3 students from each grade level.

```
stratified = df.groupby("grade", group_keys=False).apply(
    lambda x: x.sample(n=3)
)
print(f"Stratified mean: {stratified['score'].mean():.1f}")
print(stratified[["student_id", "grade", "score"]])
```

3. Cluster sample. Randomly select one entire grade level as the cluster.

```
chosen_grade = pd.Series(df["grade"].unique()).sample(n=1).values[0]
cluster = df[df["grade"] == chosen_grade]
print(f"Cluster selected: {chosen_grade}")
print(f"Cluster mean: {cluster['score'].mean():.1f}")
```

Exercise 6 Comparing Sampling Methods

Working Space

Run each of the three samples above three times.

1. Record each sample mean in the table.

Method	Run 1	Run 2	Run 3	Most stable?
SRS				
Stratified				
Cluster				
True mean	79.7			

2. Does the SRS always represent all three grades?
3. Which method gives the most stable estimate? Why does the stratified sample tend to stay closer to the true mean than the SRS?
4. Why does the cluster mean vary the most?

Answer on Page 18

1.5 Bias in Surveys

A random sample can still produce misleading results, because bias can enter after selection through how questions are asked and who responds.

The chance difference between a sample result and the population value is called *sampling error*. Different samples give different answers, and this variation shrinks as the sample grows. Bias behaves differently: it pushes results in one direction, and a larger sample does not reduce it.

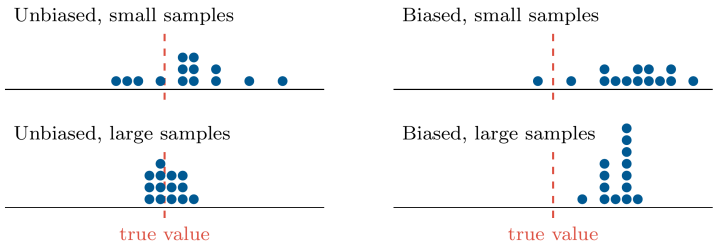


Figure 1.12: Each dot is the estimate from one sample. Sampling error shrinks as the sample size grows. Bias moves every estimate away from the true value, whatever the sample size.

Four types of bias are common in surveys.

response bias Respondents give answers that differ from their true beliefs or behaviors, often because the topic is sensitive or an interviewer creates social pressure. Food-preparation workers asked about hand-washing by a hygiene inspector will likely overstate how often they wash, because they know the answer the inspector wants.

nonresponse bias People selected for the sample cannot be reached or decline to take part, and they differ systematically from those who respond. Engaged parents return a mailed school-satisfaction survey at higher rates than disengaged parents, so the results overstate satisfaction.

undercoverage bias Part of the population is missing from the sampling frame. A survey about internet access run through online banner ads cannot reach people without internet access, the group most relevant to the question.

wording effect bias The phrasing of a question shifts the answer. Compare these two versions:

- "Do you support stricter emissions standards for vehicles?"
- "Do you support stricter emissions standards for vehicles, even if it means higher car prices and fewer model choices?"

Attaching costs to the second version makes "no" more likely, even among people who support the policy.

Exercise 7 Diagnosing Bias

Identify the most likely type of bias in each scenario and explain why it applies.

1. A streaming platform sends a satisfaction survey to all subscribers but receives responses from only 8% of them. The platform reports that 91% of subscribers are satisfied.
2. A survey on recycling habits is conducted by going door-to-door in affluent neighborhoods only, because those streets are easier and safer to walk.
3. An interviewer asks employees: "Our new open-plan office is designed to encourage collaboration. How much has it improved your productivity?"
4. A survey on alcohol consumption is conducted at a community health fair by a team wearing hospital uniforms. Respondents significantly underreport their weekly intake.

Working Space

Answer on Page 19

Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 2)

1. The population is all households in the city.
2. The frame is the city's water billing records.
3. The sample is the 200 households selected and interviewed.
4. No. A census would require interviewing every household in the city. Only 200 were selected here.

Answer to Exercise 2 (on page 4)

- Library volunteers: Census. With only 8 volunteers, asking all of them takes almost no effort.
- Battery lifespan: Sample. Testing can destroy the batteries, so a census would leave nothing to sell.
- Commute times: Sample. The population of 40 million is far too large for a full census to be practical.
- Dietary restrictions: Census. With only 22 guests, collecting complete information is straightforward and important for planning.

Answer to Exercise 3 (on page 8)

1. Number the 300 drivers from 001 to 300. Enter a random number table at any point and read three-digit numbers. Select any driver whose number falls between 001 and 300, skipping values from 301 to 999 and any numbers that have already been selected, until 25 drivers have been chosen. Alternatively, use `randInt(1,300,25)` on a TI-84, ignoring repeats.
2. This is a convenience sample. Drivers who are easy to reach at shift changes may work particular routes or shifts that differ systematically from others. For example,

they may have different safety exposure than drivers on overnight or rural routes.

3. No. An SRS requires that every possible group of 25 drivers from all 300 has an equal chance of being chosen. This procedure cannot produce a sample where all 25 come from the same depot, so not all groups of 25 are possible. This is a stratified sample, not an SRS.

Answer to Exercise 4 (on page 9)

1. Answers will vary. The five sample means differ from each other and from the true mean, because each SRS contains a different set of entries. A sample that happens to include one or two very large entries lands far above the true mean.
2. With $n = 80$, the sample means cluster more closely around the true mean. Sampling error shrinks as the sample size grows.

Answer to Exercise 5 (on page 12)

- Supermarket: Cluster sampling. The stores are clusters, which means entire clusters are included once selected.
- University: Stratified random sampling. The four faculties are strata. An SRS is taken from each.
- Water utility: Systematic sampling. Random start followed by every 25th unit.
- Film festival: Simple random sample. Every possible group of 90 attendees is equally likely.

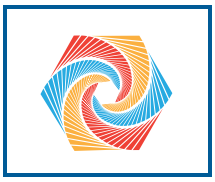
Answer to Exercise 6 (on page 14)

1. Answers will vary. Stratified means stay within a few points of 79.7. Each cluster mean is exactly 70.3, 76.5, or 92.4, depending on the grade selected.
2. No. An SRS of 9 can oversample one grade and undersample another. Because 11th graders score much higher than 9th graders, an SRS heavy in 11th graders overestimates the true mean.
3. The stratified sample. Every grade is always represented, and scores vary less within a grade than across the whole school, so the estimate stays close to the true mean from run to run.

4. The cluster sample is a single grade, so its mean depends entirely on which grade is chosen. The grades differ in score, so the cluster mean can land far from 79.7.

Answer to Exercise 7 (on page 16)

1. Nonresponse bias. The 92% who did not respond are unlikely to be satisfied in the same proportions as those who did. Highly satisfied and highly dissatisfied subscribers are both more motivated to respond than those with neutral feelings, but the pattern of non-response is unknown and the reported figure cannot be trusted.
2. Undercoverage bias. Residents of lower-income neighborhoods are excluded from the frame entirely. Recycling behavior may differ systematically by neighborhood type, so the results cannot generalize to the full population.
3. Wording effect bias. The question assumes the office has improved productivity and asks only "how much." Respondents are not given a neutral opportunity to say productivity has not changed or has declined.
4. Response bias. The formal appearance of the interviewers creates social pressure to give a health-conscious answer. Respondents adjust their reported behavior to match what they believe the interviewers expect.



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